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Отчёт к лабораторной работе №5 по предмету «Администрирование в ОС Linux» «Управление ресурсами и изоляция с помощью cgroups, namespaces. Контейнеры»

1) Создан пользователь user-54 и для его cgroup установлена квота процессора 50%

```
[30-03-2026 10:38:12] {66381ca2} student@vm-amd:~$ sudo useradd -m user-54
[sudo] password for student:
[30-03-2026 10:42:42] {66381ca2} student@vm-amd:~$ sudo systemctl set-property user-54.slice CPUQuota=50%
[30-03-2026 10:42:50] {66381ca2} student@vm-amd:~$ systemctl show user-54.slice -p CPUQuotaPerSecUSec
CPUQuotaPerSecUSec=500ms
[30-03-2026 10:42:58] {66381ca2} student@vm-amd:~$ sudo cat /sys/fs/cgroup/user.slice/user-54.slice/cpu.max
50000 100000
```

2) Для созданной cgroup установлен лимит памяти 1040 МБ и запущена нагрузка stress с потреблением 1300 МБ. Подтверждено срабатывание ограничения памяти и принудительное прерывание процесса при исчерпании лимита

```
[30-03-2026 10:43:22] {66381ca2} student@vm-amd:~$ sudo mkdir -p /sys/fs/cgroup/lab5_mem_54
[30-03-2026 10:49:15] {66381ca2} student@vm-amd:~$ echo $((1040*1024*1024)) | sudo tee /sys/fs/cgroup/lab5_mem_54/m
emory.max
1090519040
[30-03-2026 10:49:26] {66381ca2} student@vm-amd:~$ tail /dev/zero >/dev/null &
[1] 1277
[30-03-2026 10:49:32] {66381ca2} student@vm-amd:~$ PID=$!
[30-03-2026 10:49:40] {66381ca2} student@vm-amd:~$ echo "PID=$PID"
PID=1277
[30-03-2026 10:49:45] {66381ca2} student@vm-amd:~$ echo $PID | sudo tee /sys/fs/cgroup/lab5_mem_54/cgroup.procs
1277
tee: /sys/fs/cgroup/lab5_mem_54/cgroup.procs: No such process
[1]+  Killed                  tail /dev/zero > /dev/null
[30-03-2026 10:50:02] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_mem_54/memory.max
1090519040
[30-03-2026 10:50:10] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_mem_54/memory.current
0
[30-03-2026 11:01:39] {66381ca2} student@vm-amd:~$ sudo sh -c 'echo $$ > /sys/fs/cgroup/lab5_mem_54/cgroup.procs; s
tress --vm 1 --vm-bytes 1300M --vm-keep -t 20'
stress: info: [1743] dispatching hogs: 0 cpu, 0 io, 1 vm, 0 hdd
stress: FAIL: [1743] (425) <-- worker 1744 got signal 9
stress: WARN: [1743] (427) now reaping child worker processes
stress: FAIL: [1743] (431) kill error: No such process
stress: FAIL: [1743] (461) failed run completed in 5s
[30-03-2026 11:02:06] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_mem_54/memory.events
low 0
high 0
max 37
oom 1
oom_kill 1
oom_group_kill 0
```

3) Для cgroup lab5_backup_54 настроены ограничения дискового ввода-вывода riops=1540, wiops=1040 . Скрипт backup.sh запущен через cgexec внутри этой группы, а файл io.stat подтвердил выполнение операций чтения/записи в ограниченной cgroup

```
[30-03-2026 11:43:33] {66381ca2} student@vm-amd:~$ sudo cgcreate -g io:/lab5_backup_54
[30-03-2026 11:46:05] {66381ca2} student@vm-amd:~$ DEV=$(lsblk -no MAJ:MIN "$(findmnt -no SOURCE /)")
[30-03-2026 11:46:10] {66381ca2} student@vm-amd:~$ echo "$DEV"
  8:2
[30-03-2026 11:46:17] {66381ca2} student@vm-amd:~$ echo "$DEV riops=1540 wiops=1040" | sudo tee /sys/fs/cgroup/lab5
_backup_54/io.max
  8:2   riops=1540 wiops=1040
tee: /sys/fs/cgroup/lab5_backup_54/io.max: No such device
[30-03-2026 11:46:27] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_backup_54/io.max
[30-03-2026 11:46:33] {66381ca2} student@vm-amd:~$ cat > backup.sh <<'EOF'
> dd if=/dev/zero of=/tmp/backup_test.img bs=1M count=1024 oflag=direct status=progress
dd if=/tmp/backup_test.img of=/dev/null bs=1M iflag=direct status=progress
EOF
chmod +x backup.sh
[30-03-2026 11:46:52] {66381ca2} student@vm-amd:~$ sudo cgexec -g io:lab5_backup_54 ./backup.sh
919601152 bytes (920 MB, 877 MiB) copied, 1 s, 917 MB/s
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 1.23421 s, 870 MB/s
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 0.722915 s, 1.5 GB/s
[30-03-2026 11:47:01] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_backup_54/io.stat
8:0 rbytes=1073848320 wbytes=1073741824 rios=1034 wios=1024 dbytes=0 dios=0
```

```
[30-03-2026 11:47:07] {66381ca2} student@vm-amd:~$ echo "8:0 riops=1540 wiops=1040" | sudo tee /sys/fs/cgroup/lab5_
backup_54/io.max
8:0 riops=1540 wiops=1040
[30-03-2026 11:50:35] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_backup_54/io.max
8:0 rbps=max wbps=max riops=1540 wiops=1040
[30-03-2026 11:50:42] {66381ca2} student@vm-amd:~$ sudo cgexec -g io:lab5_backup_54 ./backup.sh
1048576000 bytes (1.0 GB, 1000 MiB) copied, 2 s, 524 MB/s
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 2.0435 s, 525 MB/s
817889280 bytes (818 MB, 780 MiB) copied, 1 s, 818 MB/s
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 1.37097 s, 783 MB/s
[30-03-2026 11:50:50] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_backup_54/io.stat
8:0 rbytes=2147627008 wbytes=2147483648 rios=3091 wios=3072 dbytes=0 dios=0
```

4) Для процесса top создана и настроена cgroup lab5_cpu0_54 с cpuset.cpus=0 и cpuset.mems=0. Процесс перемещён в эту группу через cgroup.procs. Проверка taskset -p показала маску 1, что подтверждает закрепление процесса только за CPU0

```
[30-03-2026 11:50:57] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/cgroup.controllers
cpuset cpu io memory hugetlb pids rdma misc
[30-03-2026 11:54:47] {66381ca2} student@vm-amd:~$ echo +cpuset | sudo tee /sys/fs/cgroup/cgroup.subtree_control
+cpuset
[30-03-2026 11:54:54] {66381ca2} student@vm-amd:~$ sudo mkdir -p /sys/fs/cgroup/lab5_cpu0_54
[30-03-2026 11:54:59] {66381ca2} student@vm-amd:~$ echo 0 | sudo tee /sys/fs/cgroup/lab5_cpu0_54/cpuset.cpus
0
[30-03-2026 11:55:06] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/cpuset.mems | sudo tee /sys/fs/cgroup/lab5_cp
u0_54/cpuset.mems
cat: /sys/fs/cgroup/cpuset.mems: No such file or directory
[30-03-2026 11:55:17] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/cpuset.mems.effective
echo 0 | sudo tee /sys/fs/cgroup/lab5_cpu0_54/cpuset.mems
0
0
[30-03-2026 11:58:18] {66381ca2} student@vm-amd:~$ top -b -d 1 >/dev/null &
[1] 2283
[30-03-2026 11:58:27] {66381ca2} student@vm-amd:~$ PID=$!
[30-03-2026 11:58:34] {66381ca2} student@vm-amd:~$ echo $PID | sudo tee /sys/fs/cgroup/lab5_cpu0_54/cgroup.procs
2283
[30-03-2026 11:58:39] {66381ca2} student@vm-amd:~$ taskset -p $PID
pid 2283's current affinity mask: 1
[30-03-2026 11:58:44] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_cpu0_54/cpuset.cpus
0
[30-03-2026 11:58:49] {66381ca2} student@vm-amd:~$ cat /sys/fs/cgroup/lab5_cpu0_54/cpuset.mems
0
```

5) Разработан и запущен сценарий, который принимает PID процесса и динамически изменяет cpu.max в cgroup в зависимости от общей загрузки CPU. Переключение режимов подтверждено

```
[30-03-2026 12:07:44] {66381ca2} student@vm-amd:~$ cat > dynamic_cpu.sh <<'EOF'
> #!/usr/bin/env bash
set -euo pipefail

if [ $# -ne 1 ]; then
  echo "Usage: $0 <PID>"
  exit 1
fi

PID="$1"
CG="/sys/fs/cgroup/lab5_dyn_54"

if ! kill -0 "$PID" 2>/dev/null; then
  echo "PID $PID not found"
  exit 1
fi

if ! grep -qw cpu /sys/fs/cgroup/cgroup.subtree_control; then
  echo +cpu | sudo tee /sys/fs/cgroup/cgroup.subtree_control >/dev/null || true
fi

sudo mkdir -p "$CG"
echo "$PID" | sudo tee "$CG/cgroup.procs" >/dev/null

echo "50000 100000" | sudo tee "$CG/cpu.max" >/dev/null

get_cpu_usage() {
  read -r _ user nice system idle iowait irq softirq steal _ _ < /proc/stat
  total1=$((user+nice+system+idle+iowait+irq+softirq+steal))
  idle1=$((idle+iowait))
  sleep 1
  read -r _ user nice system idle iowait irq softirq steal _ _ < /proc/stat
  total2=$((user+nice+system+idle+iowait+irq+softirq+steal))
  idle2=$((idle+iowait))
  dt=$((total2-total1))
  di=$((idle2-idle1))
  usage=$(( (100*(dt-di))/dt ))
  echo "$usage"
}

while true; do
  usage=$(get_cpu_usage)
```

```
  if [ "$usage" -lt 20 ]; then

    echo "80000 100000" | sudo tee "$CG/cpu.max" >/dev/null
    mode="LOW(<20):80%"
  elif [ "$usage" -gt 60 ]; then

    echo "30000 100000" | sudo tee "$CG/cpu.max" >/dev/null
mode="HIGH(>60):30%"
> else
    echo "50000 100000" | sudo tee "$CG/cpu.max" >/dev/null
    mode="MID(20-60):50%"
  fi
> cur=$(cat "$CG/cpu.max")
  echo "$(date '+%H:%M:%S') CPU=${usage}% mode=${mode} cpu.max='${cur}' pid=${PID}"
  sleep 2
done
> EOF
[30-03-2026 12:11:20] {66381ca2} student@vm-amd:~$ chmod +x dynamic_cpu.sh
[30-03-2026 12:11:42] {66381ca2} student@vm-amd:~$ yes > /dev/null &
TPID=$!
echo $TPID
[2] 2313
2313
[30-03-2026 12:11:49] {66381ca2} student@vm-amd:~$ ./dynamic_cpu.sh $TPID
12:11:57 CPU=22% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:00 CPU=9% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:03 CPU=30% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:06 CPU=9% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:09 CPU=29% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:12 CPU=11% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:15 CPU=32% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:18 CPU=11% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:21 CPU=37% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:24 CPU=9% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:28 CPU=34% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:31 CPU=9% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:34 CPU=26% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:37 CPU=9% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:40 CPU=32% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:43 CPU=24% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:46 CPU=10% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:49 CPU=100% mode=HIGH(>60):30% cpu.max='30000 100000' pid=2313
12:12:52 CPU=5% mode=LOW(<20):80% cpu.max='80000 100000' pid=2313
12:12:55 CPU=37% mode=MID(20-60):50% cpu.max='50000 100000' pid=2313
12:12:59 CPU=100% mode=HIGH(>60):30% cpu.max='30000 100000' pid=2313
12:13:02 CPU=100% mode=HIGH(>60):30% cpu.max='30000 100000' pid=2313
12:13:05 CPU=100% mode=HIGH(>60):30% cpu.max='30000 100000' pid=2313
12:13:08 CPU=100% mode=HIGH(>60):30% cpu.max='30000 100000' pid=2313
```

```
[30-03-2026 12:12:51] {66381ca2} student@vm-amd:~$ stress --cpu 4
stress: info: [2599] dispatching hogs: 4 cpu, 0 io, 0 vm, 0 hdd
stress: info: [2599] successful run completed in 20s
```

6) Запущена оболочка в отдельном UTS namespace. Внутри изолированного пространства имя хоста изменено на isolated-student-54. В отдельном терминале хоста имя осталось исходным (vm-amd), что подтверждает изоляцию UTS namespace

```
[30-03-2026 12:13:32] {66381ca2} student@vm-amd:~$ sudo unshare --uts --fork --mount-proc bash
[30-03-2026 12:19:54] {66381ca2} root@vm-amd:/home/student# hostname isolated-student-54
[30-03-2026 12:20:02] {66381ca2} root@vm-amd:/home/student# hostname
isolated-student-54
```

```
[30-03-2026 12:19:39] {66381ca2} student@vm-amd:~$ hostname
vm-amd
```

7) Создано отдельное PID namespace, внутри которого после монтирования proc команда ps aux отображала только локальные процессы пространства имен. В отдельной хостовой сессии выведен полный список системных процессов, что подтверждает изоляцию PID namespace

```
[30-03-2026 15:54:12] {40ab33fa} student@vm-amd:~$ sudo unshare --pid --fork bash
[sudo] password for student:
[30-03-2026 15:54:43] {40ab33fa} root@vm-amd:/home/student# mount -t proc proc /proc
[30-03-2026 15:54:50] {40ab33fa} root@vm-amd:/home/student# ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.2   7604  4524 pts/1    S    15:54   0:00 bash
root          11  0.0  0.2  10884  4572 pts/1    R+   15:54   0:00 ps aux
```

```
[30-03-2026 16:02:53] {40ab33fa} student@vm-amd:~$ ps aux | head -n 20
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.2  0.6  21976 13248 ?        Ss   15:53   0:01 /sbin/init splash noprompt nosh
ell automatic-ubiquity
root           2  0.0  0.0      0     0 ?        S    15:53   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    15:53   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   15:53   0:00 [kworker/R-rcu_g]
root           5  0.0  0.0      0     0 ?        I<   15:53   0:00 [kworker/R-rcu_p]
root           6  0.0  0.0      0     0 ?        I<   15:53   0:00 [kworker/R-slub_]
root           7  0.0  0.0      0     0 ?        I<   15:53   0:00 [kworker/R-netns]
root           9  0.1  0.0      0     0 ?        I    15:53   0:00 [kworker/0:1-cgroup_release]
root          11  0.0  0.0      0     0 ?        I    15:53   0:00 [kworker/u2:0-events_power_effi
cient]
root          12  0.0  0.0      0     0 ?        I<   15:53   0:00 [kworker/R-mm_pe]
root          13  0.0  0.0      0     0 ?        I    15:53   0:00 [rcu_tasks_kthread]
root          14  0.0  0.0      0     0 ?        I    15:53   0:00 [rcu_tasks_rude_kthread]
root          15  0.0  0.0      0     0 ?        I    15:53   0:00 [rcu_tasks_trace_kthread]
root          16  0.0  0.0      0     0 ?        S    15:53   0:00 [ksoftirqd/0]
root          17  0.0  0.0      0     0 ?        I    15:53   0:00 [rcu_preempt]
root          18  0.0  0.0      0     0 ?        S    15:53   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    15:53   0:00 [idle_inject/0]
root          20  0.0  0.0      0     0 ?        S    15:53   0:00 [cpuhp/0]
root          21  0.0  0.0      0     0 ?        S    15:53   0:00 [kdevtmpfs]
```

8) В отдельном mount namespace создан каталог /tmp/private_root и смонтирован tmpfs. Внутри namespace мы видим точку монтирования (/tmp/private_root), тогда как в хостовой сессии вывод отсутствует, что подтверждает изоляцию mount namespace

```
[30-03-2026 16:04:52] {40ab33fa} root@vm-amd:/home/student# sudo unshare --mount bash
[30-03-2026 16:06:29] {40ab33fa} root@vm-amd:/home/student# mkdir -p /tmp/private_$(whoami)
[30-03-2026 16:06:34] {40ab33fa} root@vm-amd:/home/student# mount -t tmpfs tmpfs /tmp/private_$(wh
oami)
[30-03-2026 16:06:39] {40ab33fa} root@vm-amd:/home/student# df -h | grep private_$(whoami)
tmpfs           985M     0  985M   0% /tmp/private_root
```

```
[30-03-2026 16:03:02] {40ab33fa} student@vm-amd:~$ df -h | grep private_$(whoami)
[30-03-2026 16:07:12] {40ab33fa} student@vm-amd:~$
```

9) Запущена оболочка в отдельном network namespace. Внутри пространства имен команда ip addr показала только интерфейс lo, а ping google.com завершился ошибкой разрешения имени, что подтверждает отсутствие сетевого доступа

```
[30-03-2026 16:06:45] {40ab33fa} root@vm-amd:/home/student# sudo unshare --net bash
[30-03-2026 17:22:16] {40ab33fa} root@vm-amd:/home/student# ip addr
1: lo: <LOOPBACK> mtu 65536 qdisc noop state DOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
[30-03-2026 17:22:27] {40ab33fa} root@vm-amd:/home/student# ping -c 3 google.com
ping: google.com: Temporary failure in name resolution
```

```
[30-03-2026 17:23:05] {40ab33fa} student@vm-amd:~$ ip addr
ping -c 3 8.8.8.8
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1
000
    link/ether 08:00:27:e0:36:cf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 80872sec preferred_lft 80872sec
    inet6 fd17:625c:f037:2:a00:27ff:fee0:36cf/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86151sec preferred_lft 14151sec
    inet6 fe80::a00:27ff:fee0:36cf/64 scope link
       valid_lft forever preferred_lft forever
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2073ms
```

10a) Выполнена начальная настройка OverlayFS: созданы каталоги lower, upper, work, merged в ~/overlay_54, в нижнем слое создан файл 54_original.txt с текстом «Оригинальный текст из LOWER»

```
[30-03-2026 17:26:23] {40ab33fa} student@vm-amd:~$ mkdir -p ~/overlay_54/{lower,upper,work,merged}
[30-03-2026 18:00:19] {40ab33fa} student@vm-amd:~$ echo "Оригинальный текст из LOWER" > ~/overlay_54/lower/54_origi
nal.txt
[30-03-2026 18:00:26] {40ab33fa} student@vm-amd:~$ sudo mount -t overlay overlay \
  -o lowerdir=$HOME/overlay_54/lower,upperdir=$HOME/overlay_54/upper,workdir=$HOME/overlay_54/work \
  $HOME/overlay_54/merged
[sudo] password for student:
[30-03-2026 18:01:02] {40ab33fa} student@vm-amd:~$ ls -la ~/overlay_54/lower
total 12
drwxrwxr-x 2 student student 4096 Mar 30 18:00 .
drwxrwxr-x 6 student student 4096 Mar 30 18:00 ..
-rw-rw-r-- 1 student student   47 Mar 30 18:00 54_original.txt
[30-03-2026 18:01:17] {40ab33fa} student@vm-amd:~$ ls -la ~/overlay_54/merged
total 12
drwxrwxr-x 1 student student 4096 Mar 30 18:00 .
drwxrwxr-x 6 student student 4096 Mar 30 18:00 ..
-rw-rw-r-- 1 student student   47 Mar 30 18:00 54_original.txt
[30-03-2026 18:01:21] {40ab33fa} student@vm-amd:~$ mount | grep overlay_54
overlay on /home/student/overlay_54/merged type overlay (rw,relatime,lowerdir=/home/student/overlay_54/lower,upperd
ir=/home/student/overlay_54/upper,workdir=/home/student/overlay_54/work,uuid=on,nouserxattr)
```

10b)В объединённом слое merged удалён файл 54_original.txt, после чего в верхнем слое upper появился whiteout-артефакт

```
[30-03-2026 18:01:25] {40ab33fa} student@vm-amd:~$ rm ~/overlay_54/merged/54_original.txt
[30-03-2026 18:07:49] {40ab33fa} student@vm-amd:~$ ls -la ~/overlay_54/upper
total 8
drwxrwxr-x 2 student student 4096 Mar 30 18:07 .
drwxrwxr-x 6 student student 4096 Mar 30 18:00 ..
c--------- 2 root    root    0, 0 Mar 30 18:07 54_original.txt
[30-03-2026 18:07:55] {40ab33fa} student@vm-amd:~$ ls -la ~/overlay_54/upper | cat -A
total 8$
drwxrwxr-x 2 student student 4096 Mar 30 18:07 .$
drwxrwxr-x 6 student student 4096 Mar 30 18:00 ..$
c--------- 2 root    root    0, 0 Mar 30 18:07 54_original.txt$
[30-03-2026 18:08:05] {40ab33fa} student@vm-amd:~$ rm -f ~/overlay_54/upper/.wh.54_original.txt
[30-03-2026 18:08:20] {40ab33fa} student@vm-amd:~$ ls -la ~/overlay_54/merged
total 8
drwxrwxr-x 1 student student 4096 Mar 30 18:07 .
drwxrwxr-x 6 student student 4096 Mar 30 18:00 ..
[30-03-2026 18:08:27] {40ab33fa} student@vm-amd:~$ cat ~/overlay_54/merged/54_original.txt
cat: /home/student/overlay_54/merged/54_original.txt: No such file or directory
```

```
[30-03-2026 18:16:25] {40ab33fa} student@vm-amd:~$ ls -la ~/overlay_54/upper
total 8
drwxrwxr-x 2 student student 4096 Mar 30 18:12 .
drwxrwxr-x 6 student student 4096 Mar 30 18:00 ..
[30-03-2026 18:16:30] {40ab33fa} student@vm-amd:~$ ls -la ~/overlay_54/merged
total 12
drwxrwxr-x 1 student student 4096 Mar 30 18:12 .
drwxrwxr-x 6 student student 4096 Mar 30 18:00 ..
-rw-rw-r-- 1 student student   47 Mar 30 18:00 54_original.txt
[30-03-2026 18:16:37] {40ab33fa} student@vm-amd:~$ cat ~/overlay_54/merged/54_original.txt
Оригинальный текст из LOWER
```

10c) Разработан скрипт overlay_audit.sh, формирующий отчёт 54_audit.log по состоянию OverlayFS. Скрипт выполняет: поиск whiteout-объектов в upper, сравнение списков файлов в lower и merged, выявление отсутствующих файлов и расхождений содержимого

```
[30-03-2026 18:22:10] {40ab33fa} student@vm-amd:~$ cat > ~/overlay_54/overlay_audit.sh <<'EOF'
> #!/usr/bin/env bash
set -euo pipefail

ID=54
BASE="$HOME/overlay_${ID}"
LOWER="$BASE/lower"
UPPER="$BASE/upper"
MERGED="$BASE/merged"
LOG="$BASE/${ID}_audit.log"
> {
> echo "OverlayFS audit for ID=$ID"
  echo "Timestamp: $(date)"
  echo

  echo "[1] Whiteout entries in upper:"
  # whiteout как char device 0,0 и/или .wh.* файлы
  find "$UPPER" -xdev \( -type c -o -name '.wh.*' \) -print || true
  echo

  echo "[2] Files in LOWER:"
  find "$LOWER" -type f | sed "s#^$LOWER/##" | sort
  echo

  echo "[3] Files in MERGED:"
  find "$MERGED" -type f | sed "s#^$MERGED/##" | sort
  echo
> echo "[4] Missing in MERGED (present in LOWER):"
  while IFS= read -r rel; do
    [ -z "$rel" ] && continue
    if [ ! -e "$MERGED/$rel" ]; then
      echo "MISSING: $rel"
    fi
  done < <(find "$LOWER" -type f | sed "s#^$LOWER/##" | sort)
  echo
> echo "[5] Content mismatches (LOWER vs MERGED):"
  while IFS= read -r rel; do
    [ -z "$rel" ] && continue
    if [ -e "$MERGED/$rel" ] && ! cmp -s "$LOWER/$rel" "$MERGED/$rel"; then
      echo "DIFF: $rel"
    fi
> done < <(find "$LOWER" -type f | sed "s#^$LOWER/##" | sort)
} | tee "$LOG"
> EOF
```

```
[30-03-2026 18:23:10] {40ab33fa} student@vm-amd:~$ chmod +x ~/overlay_54/overlay_audit.sh
[30-03-2026 18:23:17] {40ab33fa} student@vm-amd:~$ ~/overlay_54/overlay_audit.sh
OverlayFS audit for ID=54
Timestamp: Mon Mar 30 06:23:22 PM MSK 2026

[1] Whiteout entries in upper:

[2] Files in LOWER:
54_original.txt

[3] Files in MERGED:
54_original.txt

[4] Missing in MERGED (present in LOWER):

[5] Content mismatches (LOWER vs MERGED):
[30-03-2026 18:23:22] {40ab33fa} student@vm-amd:~$ cat ~/overlay_54/54_audit.log
OverlayFS audit for ID=54
Timestamp: Mon Mar 30 06:23:22 PM MSK 2026

[1] Whiteout entries in upper:

[2] Files in LOWER:
54_original.txt

[3] Files in MERGED:
54_original.txt

[4] Missing in MERGED (present in LOWER):

[5] Content mismatches (LOWER vs MERGED):
```

10d)

1) **Как OverlayFS скрывает файлы из lower при удалении в merged?**
   OverlayFS не удаляет файл из lower, а создаёт в upper специальную запись whiteout, которая маскирует соответствующий файл нижнего слоя в объединённом представлении
2) **Если удалить каталог work, можно ли перемонтировать overlay? Почему?**
   Нет, потому что ядро ведёт внутренние операции через workdir (без валидного workdir mount вернёт ошибку)
3) **Что будет с merged, если upper пустой?**
   Будет отображать содержимое lower практически как есть: видны файлы нижнего слоя, а изменений/whiteout из верхнего слоя нет

11) Подготовлен оптимизированный Docker-образ Flask-приложения: использован фиксированный lightweight-базовый образ, зависимости вынесены в отдельный кэшируемый слой, запуск контейнера выполнен от непривилегированного пользователя, добавлен .dockerignore

```
[30-03-2026 18:54:15] {40ab33fa} student@vm-amd:~/lab5_task11$ cat Dockerfile
FROM python:3.12.8-slim-bookworm

ENV PYTHONDONTWRITEBYTECODE=1 \
    PYTHONUNBUFFERED=1

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

RUN useradd -m -u 10001 appuser

COPY app.py .

USER appuser

EXPOSE 5000
CMD ["python", "app.py"]
[30-03-2026 18:54:31] {40ab33fa} student@vm-amd:~/lab5_task11$ cat .dockerignore
__pycache__/
*.pyc
*.pyo
*.pyd
*.log
.git
.gitignore
.env
venv/
.venv/
```

```
[30-03-2026 18:52:59] {40ab33fa} student@vm-amd:~/lab5_task11$ docker run --rm -p 5000:5000 -e STU
DENT_NAME=yaroslav lab5-flask:opt
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production
WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
```

```
[30-03-2026 18:48:41] {40ab33fa} student@vm-amd:~$ curl http://localhost:5000
Container IP: 172.17.0.2 Student: yaroslav[30-03-2026 18:53:22] {40ab33fa} student@vm-amd:~$
```

12) Создан и запущен docker-compose-стек для WordPress и MariaDB с сохранением данных в томах. Контейнеры wp-app-54 и wp-db-54 успешно подняты

```
[30-03-2026 19:34:00] {40ab33fa} student@vm-amd:~/lab5_task12$ cat docker-compose.yml
version: "3.9"
services:
  db:
    image: mariadb:11.4
    container_name: wp-db-54
    restart: unless-stopped
    environment:
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: yaroslav_db_pass
      MYSQL_ROOT_PASSWORD: yaroslav_db_pass
    volumes:
      - db-data-54:/var/lib/mysql
    healthcheck:
      test: ["CMD-SHELL", "mariadb-admin ping -h localhost -uroot -pyaroslav_db_pass || exit 1"]
      interval: 10s
      timeout: 5s
      retries: 10

  wordpress:
    image: wordpress:6.8-php8.3-apache
    container_name: wp-app-54
    restart: unless-stopped
    depends_on:
      db:
        condition: service_healthy
    ports:
      - "2054:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_NAME: wordpress
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: yaroslav_db_pass
    volumes:
      - yaroslav-wp-data:/var/www/html

volumes:
  yaroslav-wp-data:
  db-data-54:
```

```
[30-03-2026 19:34:18] {40ab33fa} student@vm-amd:~/lab5_task12$ docker-compose up -d
Creating network "lab5_task12_default" with the default driver
Creating volume "lab5_task12_yaroslav-wp-data" with default driver
Creating volume "lab5_task12_db-data-54" with default driver
Pulling db (mariadb:11.4)...
11.4: Pulling from library/mariadb
817807f3c64e: Pull complete
802f7f622100: Pull complete
c08b5b4953f7: Pull complete
c69718d5edc2: Pull complete
5cf123ffd6c6: Pull complete
8cde0a5c559a: Pull complete
9820d07392af: Pull complete
c3846618e585: Pull complete
Digest: sha256:5559e323266d86bea94f48c338014b102a2031af2895484cf9d99d58d2cb9cd3
Status: Downloaded newer image for mariadb:11.4
Pulling wordpress (wordpress:6.8-php8.3-apache)...
6.8-php8.3-apache: Pulling from library/wordpress
0e4bc2bd6656: Pull complete
5ff292d92bef: Pull complete
30d3fccd4e77: Pull complete
2e42a5718777: Pull complete
1786c40e6ef6: Pull complete
d546c1d12a7f: Pull complete
64709930f968: Pull complete
d1a33af9a2e4: Pull complete
b95356b5f483: Pull complete
8e3cf1a078fa: Pull complete
dba783666fd7: Pull complete
387c70da0b5e: Pull complete
f3c8c18e18ce: Pull complete
2da9ad8b7fb7: Pull complete
4f4fb700ef54: Pull complete
7e701ee4b7b2: Pull complete
28d867254d43: Pull complete
17988f9e24c9: Pull complete
27f4c48d5338: Pull complete
fcbc612be48f: Pull complete
8cc6cf260c6a: Pull complete
8b7bcd1f528e: Pull complete
a3a74878ba86: Pull complete
ece88f57b218: Pull complete
Digest: sha256:30bff39330d1693b0ce13d32fc9b7bb67193064f040b7d60d3494e136fa599d4
Status: Downloaded newer image for wordpress:6.8-php8.3-apache
Creating wp-db-54 ... done
Creating wp-app-54 ... done
[30-03-2026 19:35:57] {40ab33fa} student@vm-amd:~/lab5_task12$ docker-compose ps
  Name             Command               State                    Ports
-----------------------------------------------------------------------------------------
wp-app-54   docker-entrypoint.sh apach ...   Up             0.0.0.0:2054->80/tcp,:::2054->80/tcp
wp-db-54    docker-entrypoint.sh mariadbd    Up (healthy)   3306/tcp
```